

C & G Harmonic Equilibrium

Transverse-Wave Resonance and Scatter-Field Propagation

Conceptual mathematical chapter - Universal Harmonic Theory

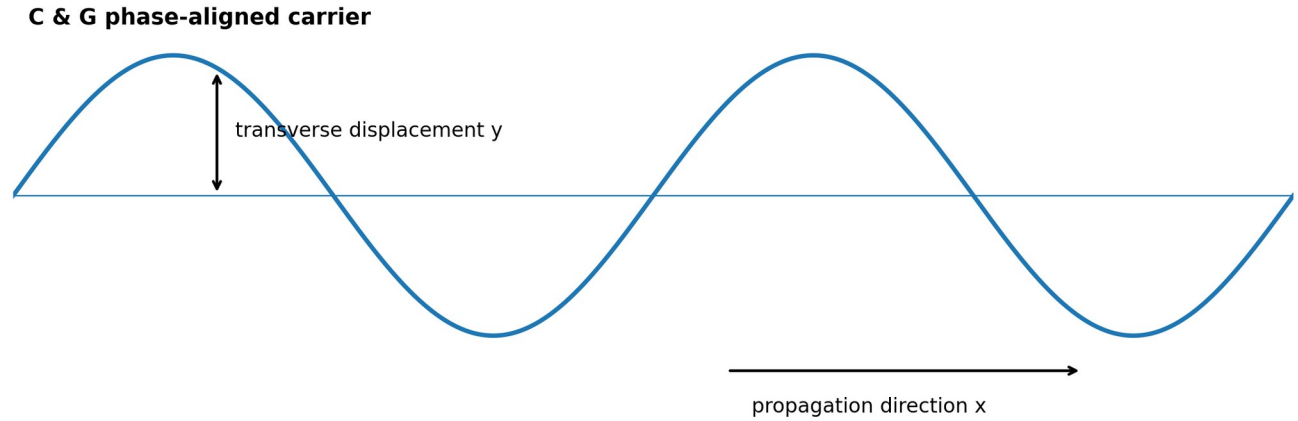


Figure 1. Transverse displacement is perpendicular to the direction of propagation.

Scientific status. The transverse-wave, resonance, interference and scattering equations below are established wave physics. The assignment of the symbols C and G as fundamental universal quantities is a proposed interpretation and must be treated as a testable hypothesis rather than an established physical law.

1. Transverse C & G carrier wave

Let C denote the coherent field component and G the geometric constraint or coupling component. Their phase-aligned transverse carrier is represented by a displacement perpendicular to propagation:

$$y_{CG}(x,t) = A_{CG} \cos(kx - \omega t + \varphi_{CG})$$

where A_{CG} is amplitude, $k = 2\pi/\lambda$ is wave number, $\omega = 2\pi f$ is angular frequency, and φ_{CG} is phase. The transverse condition is

$$y_{CG} \perp \hat{x}$$

meaning that the field displacement is perpendicular to the propagation direction.

2. Wave equation and propagation speed

An ideal one-dimensional transverse field satisfies

$$\partial^2 y_{CG} / \partial t^2 = v^2 \partial^2 y_{CG} / \partial x^2$$

with

$$v = f\lambda = \omega/k$$

For a mechanical transverse medium such as a stretched string, $v = \sqrt{T/\mu}$, where T is tension and μ is linear mass density. For an abstract field, v must be derived from the defined field equations rather than assumed.

3. Perfect resonance as an ideal limit

A mathematically ideal C & G resonance requires matched frequency, phase and polarization:

$$f_C = f_G = f_0, \quad \Delta\varphi_{CG} = \varphi_C - \varphi_G = 0, \quad \hat{e}_C = \hat{e}_G$$

A dimensionless coherence measure may be defined as

$$\Gamma_{CG} = \cos(\Delta\varphi_{CG})$$

so $\Gamma_{CG} = 1$ at exact phase alignment. A practical resonance also contains damping:

$$\partial^2 y / \partial t^2 + 2\gamma \partial y / \partial t + \omega_0^2 y = F_0 \cos(\omega t)$$

The response is largest near $\omega \approx \omega_0$, but infinite or lossless resonance is not physically attainable in a real dissipative system.

4. Perfect-fifth C:G harmonic option

Where C and G are intended to correspond to musical pitch classes C and G, their ideal just-intonation frequency ratio is the perfect fifth:

$$f_G / f_C = 3/2$$

This is a harmonic interval, not frequency equality. Therefore two distinct cases should not be confused:

- Unison resonance: $f_C = f_G$, used for exact modal coincidence.
- Perfect-fifth balance: $f_G = (3/2)f_C$, used for stable harmonic coupling between two different tones.

5. Transverse scatter-wave field

Let an incident transverse C & G carrier encounter a scattering region. The total field is

$$y_{\text{tot}}(r,t) = y_{\text{inc}}(r,t) + \sum_i y_{\text{sc},i}(r,t)$$

with an outgoing scattered component represented in the far field by

$$y_{\text{sc}}(r,t) \approx \text{Re}\{[F(\theta,\varphi)/r] \exp[i(kr - \omega t)]\}$$

where $F(\theta,\varphi)$ is the angular scattering amplitude. The observable intensity is proportional to

$$I(r) \propto |\Psi_{\text{inc}} + \Psi_{\text{sc}}|^2$$

which expands to

$$I \propto |\Psi_{\text{inc}}|^2 + |\Psi_{\text{sc}}|^2 + 2 \operatorname{Re}(\Psi_{\text{inc}} \Psi_{\text{sc}}^*)$$

The final term is the interference term. Constructive reinforcement occurs when the phase difference is near $2\pi n$; destructive cancellation occurs near $(2n+1)\pi$.

6. Resonance distillation

“Distillation” can be defined rigorously as extracting the coherent transverse mode from a multi-frequency signal. Write

$$y(t) = \sum_n A_n \cos(2\pi f_n t + \varphi_n)$$

and apply a spectral selection operator centred at f_0 :

$$y_{\text{dist}}(t) = \mathcal{F}^{-1}\{W(f - f_0) \mathcal{F}[y(t)]\}$$

where W is a narrow frequency window. A coherence-weighted version may retain components according to their phase relation to the C & G reference:

$$y_{\text{CG,dist}}(t) = \sum_n W_n \Gamma_n A_n \cos(2\pi f_n t + \varphi_n)$$

with $\Gamma_n = \cos(\varphi_n - \varphi_{\text{CG}})$. This expresses resonance distillation as frequency selection plus phase-coherence selection.

7. Compact governing formulation

A compact phenomenological equation combining transverse propagation, damping, C & G coupling and a scattering potential is

$$\partial^2 y / \partial t^2 + 2\gamma \partial y / \partial t - v^2 \nabla^2 y + \omega_0^2 y + V_{\text{sc}}(r)y = \kappa_{\text{CG}} C(r,t)G(r,t)$$

Here V_{sc} describes the scattering environment and κ_{CG} is a coupling coefficient. This equation is a proposed model form; its dimensions, boundary conditions and measurable predictions must be specified before experimental use.

8. Final definition

C & G transverse harmonic resonance is defined as a phase-organised transverse carrier in which the C and G components either coincide at one resonant frequency or maintain a specified harmonic ratio, while scattering redistributes amplitude and direction. “Perfect resonance” is the ideal limit $\Delta\varphi = 0$ and minimal damping; “resonance distillation” is the extraction of the phase-coherent spectral component from the scattered total field.